

Day 26 - Introduction to Chaos

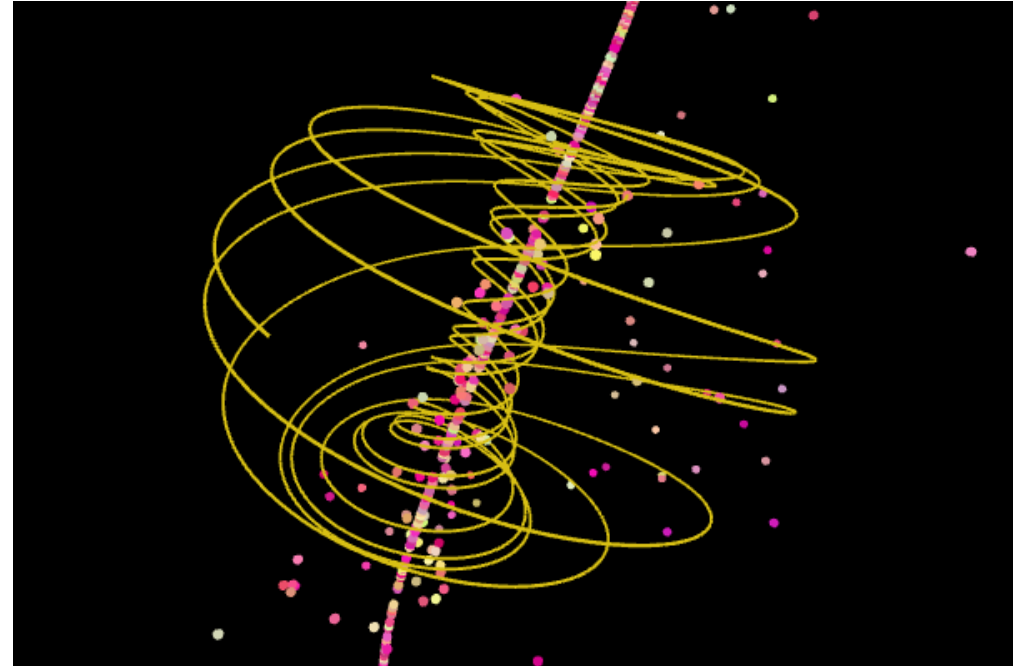
$$\dot{x} = y + axy + xz$$

$$\dot{y} = 1 - bx^2 + yz$$

$$\dot{z} = x - x^3 - y^2$$

$$a = 2.07 \quad b = 1.79$$

<https://www.dynamicmath.xyz/strange-attractors/>



Announcements

- Midterm 1 is graded
 - Feedback delivered for project
 - Still some outstanding grading issues
- Homework 7 is due Friday
 - No homework next week
- Midterm 2 will be assigned next Monday
 - Second project check-in

What is Chaos?

At your table, discuss what it means for a system to be chaotic.

- What are some examples of chaotic systems?
- What are some characteristics of chaotic systems?
- How do chaotic systems differ from non-chaotic systems?

Try to come up with two answers



Hallmarks of a Classically Chaotic System

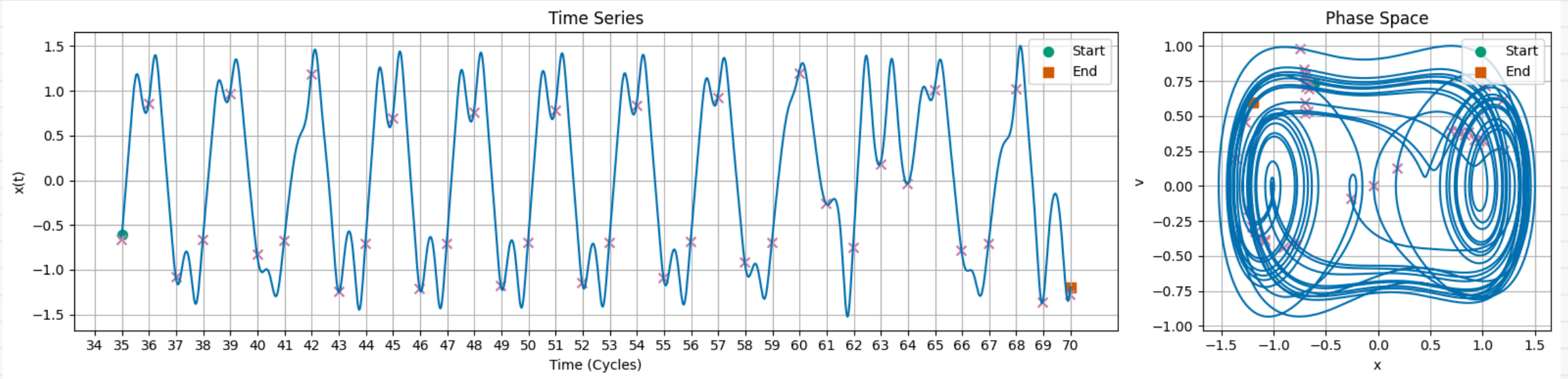
1. **Deterministic**: The system is governed by deterministic laws (e.g., Newton's laws of motion, a set of differential equations)
2. **Sensitive to Initial Conditions**: A bundle of trajectories that start close together will diverge exponentially over time
3. **Non-periodic Behavior**: The system exhibits complex, non-repeating behavior over time, this might look like "random" behavior, but it is not truly random because it is deterministic

Hallmarks of a Classically Chaotic System

4. **Strange Attractors:** The system may have a strange attractor, which is a fractal structure in phase space that the system tends to evolve towards over time
5. **Parameter Sensitivity:** The system may be sensitive to small changes in parameters, which can trigger qualitative changes in the system's behavior
6. (Sometimes) **Periodic Behavior:** The system may exhibit periodic behavior for certain parameter values, and this might be a signal that the system bifurcates into chaotic behavior for other parameter values

Example 1: Duffing Equation

$$\ddot{x} + \beta\dot{x} + \alpha x + \gamma x^3 = F_0 \cos(\omega t)$$



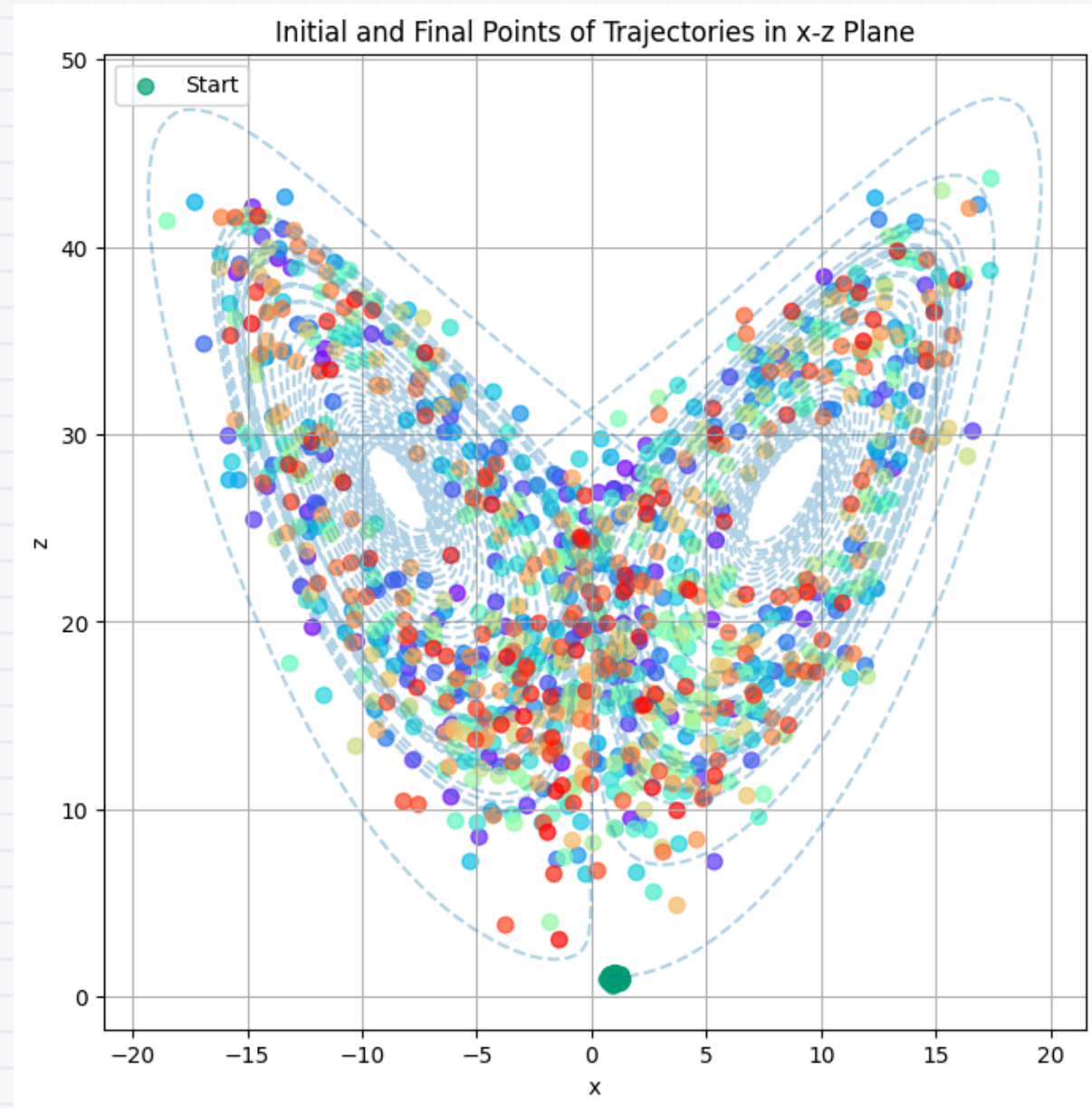
Exhibits Periodic and Chaotic Behavior

Illustrates period doubling bifurcations as route to chaos

Example 2: Lorenz System

$$\begin{aligned}\dot{x} &= \sigma(y - x) \\ \dot{y} &= x(\rho - z) - y \\ \dot{z} &= xy - \beta z\end{aligned}$$

**Exhibits sensitive dependence
on initial conditions**
**Demonstrates the concept of a
strange attractor**



Using `solve_ivp`

We will start with the damped driven pendulum as an example. This will illustrate how to use `solve_ivp` to solve a system of coupled first-order differential equations.

$$\ddot{\theta} + \beta \dot{\theta} + \sin(\theta) = A \cos(\omega_D t)$$

We can rewrite this as two first-order equations:

$$\begin{aligned}\dot{\theta} &= \omega \\ \dot{\omega} &= -\beta\omega - \sin(\theta) + A \cos(\omega_D t)\end{aligned}$$

Using `solve_ivp`

To use `solve_ivp`, we write a function for the derivatives:

```
def damped_driven_pendulum(t, y, beta, A, omegaD=1):  
    theta, omega = y  
    dtheta_dt = omega  
    domega_dt = -np.sin(theta) - beta * omega + A * np.cos(omegaD*t)  
    return [dtheta_dt, domega_dt]
```

Using `solve_ivp`

Now we can use `solve_ivp` to solve the system of equations:

```
# Parameters that define the system
beta = 0.5
A = 1.0
omegaD = 2*np.pi

# Time span for the simulation
t_span = (0, 100)

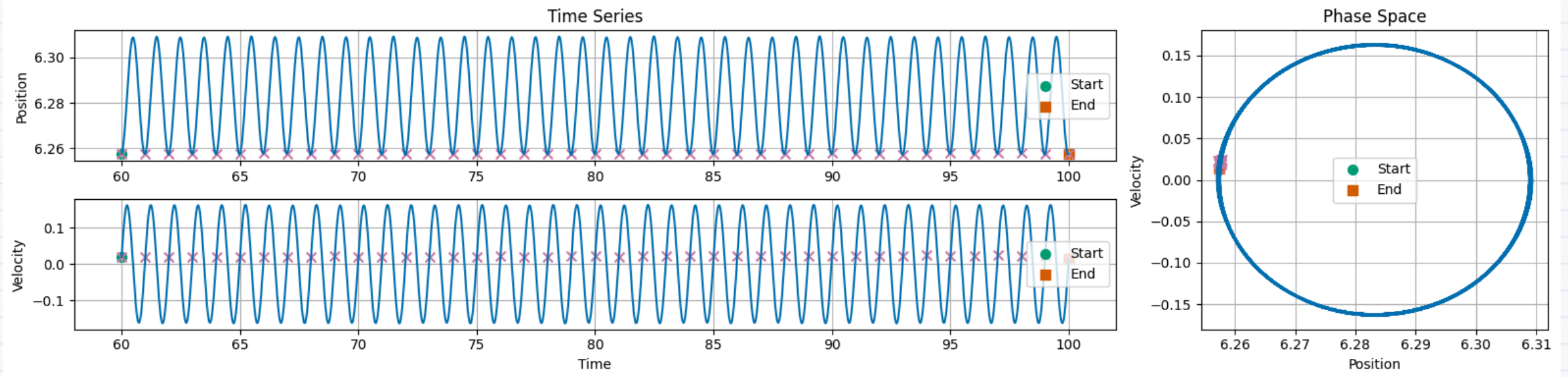
# Initial conditions: [theta, omega]
y0 = [6, 0]

# Time points where we want the solution
t_eval = np.linspace(t_span[0], t_span[1], 10000)

# Solve the system of equations
solution = solve_ivp(damped_driven_pendulum, t_span, y0, args=(beta, A, omegaD), t_eval=t_eval)
```

Damped Driven Pendulum

Long Term Behavior is Periodic



"Period-1" Dynamics is a term to indicate there's a single frequency governing the motion

Phase space plots can provide a better window into the system's behavior